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Btech CSE core.
Sec → A

Assignment → 1 DBMS Unit → 1

Sec → A Introduction to DBMS:-

Q:1→ DBMS → A DBMS is software that manages data in an organised way using table. It helps user to store update & retrieve data easily.
It is required to manage large data safely, avoid duplication, & provide secure access.

Q:2→ Why DBMS is better than file system?
DBMS stores data in a structured format & allows multiple users to access it safely.
File systems may create duplicate data & are harder to manage.

Q:3→ Data Abstraction:- It means hiding technical details & showing only required information to user.
→ Physical (how data is stored).
→ Logical (What data is stored).
→ View (what user sees.)

Q:4→ What is three-schema Architecture?
It divides database into three part-
→ Internal level.
→ Conceptual level.
→ External level.
It helps in data independence & easy modification.

5) Data Independence :- It means changing data structure are one level without affecting other. It helps in flexibility & system maintainance

6) ER Model :- It is a diagram that represents entities, their attributes, & relationship. It is used for database design planning.

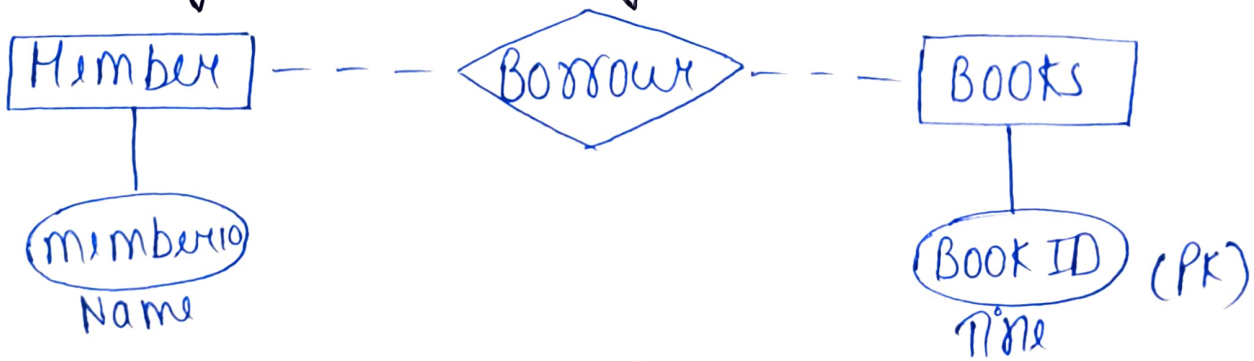
7) What are Integrity constraints :- Integrity constraints are rules that ensure correct & valid data.
Ex :- Primary Keys, foreign key, Not Null.

Sec → B Design, Draw & Defend

1→ Three Schema -

External level → user view
↓
Conceptual level → overall structure.
↓
Internal level → physical storage.

2→ ER diagram → Library



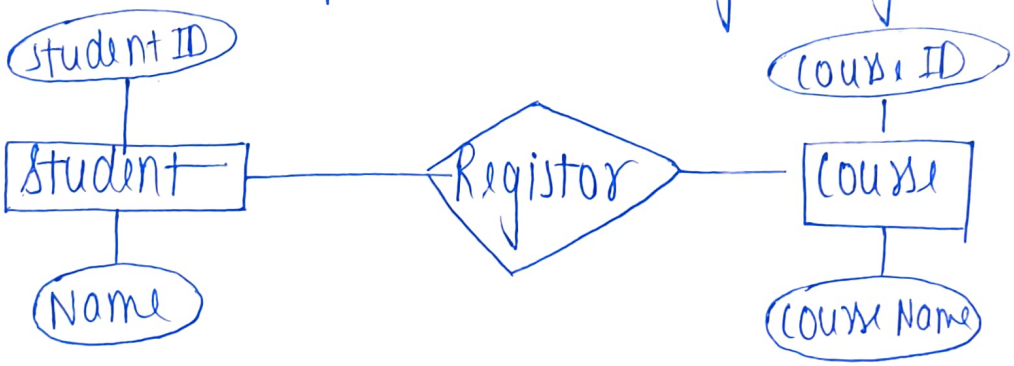
- Entities :- Member, Book.
- Relationship :- Borrow.
- PK — member ID, Book ID.

One member can borrow many books.

ii) Student - course registration ER.

- Student (Student ID, name)
- Course (Course ID, Course-Name)

Relationship :- Student registor for course.



Sec-C - Case Based Analytical Ques →

(i) ER Diagram - Hospital System

Entities → Patient, Doctor, Appointment.
Patients books Appointment
Doctor attends Appointment.

Primary Keys → Patient-ID, Doctor-ID, Appointment-ID.

ii) Online shopping :-

Customer (Customer_ID - PK)
Product (Product - ID - PK)
Order (Order - ID - PK)

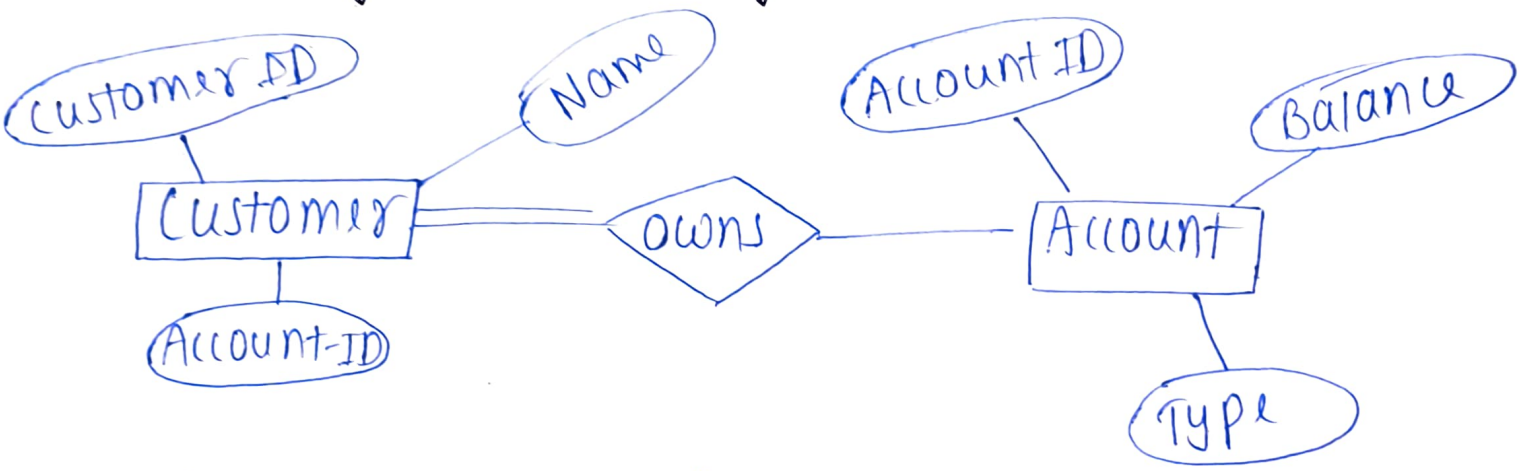
foreign key in order Table connect customer & Product

iii) University – Data Abstraction :–

Student see Marks (view level).
Admin manager rewards (logical level)
DBA · controls storage (Physical level)

Sec-D – One diagram + On Insign

(i) ER – Diagram – ATM system :–



Account – no· is primary key.
Relation :– Customer owns Account.
Account :– no take uniquely Indentifier Account.

ii) Three – Schema Architecture + One Benefit

Benefit :– Provide Data independence.
change is storage do not affect uses.

Diagram → External level (user view)
↓
conceptual level (logical structure of database)
↓
Internal level (Physical storage level)

iii) Draw a Entry with Attributes:-

Entry :- Employee

Employee - ID (PK)
Name
Salary
Department

Primary Key - Employee - ID
because for all employee their ID will be different.